

Day 25 Problem Set

Geometry Summer Credit | Mixed Review Set 1

Name:	Date:	Period:
-------	-------	---------

Directions

Choose the tool first. Show enough work to make your strategy clear.

Part A: Transformations, Congruence, and Similarity

1.	Translate A(-3, 8) by $(x+6, y-10)$.
2.	Reflect B(7, -2) across the y-axis.
3.	Rotate C(5, -4) 180 degrees about the origin.
4.	Dilate D(-6, 9) by scale factor $\frac{1}{3}$ about the origin.
5.	Two similar rectangles have scale factor 4 from smaller to larger. The smaller perimeter is 18. Find the larger perimeter.
6.	Two similar figures have scale factor 3 from smaller to larger. The smaller area is 22 square units. Find the larger area.
7.	Triangle ABC is similar to triangle DEF. $AB=10$, $DE=25$, and $AC=14$. Find DF.
8.	Triangle PQR is congruent to triangle XYZ. If angle Q=64 degrees, which angle in triangle XYZ is 64 degrees?

- | | |
|----|---|
| 9. | Error analysis: A student says a dilation by scale factor 2 creates a congruent image. Explain the error. |
|----|---|

Part B: Right-Triangle Review

10. Find the hypotenuse of a right triangle with legs 8 and 15.

11. Find the missing leg if the hypotenuse is 13 and one leg is 5.

12. Simplify $\sqrt{72}$.

13. In a 45-45-90 triangle, one leg is 11. Find the hypotenuse.

14. In a 30-60-90 triangle, the short leg is 7. Find the hypotenuse and long leg.

15. A right triangle has angle 40 degrees and adjacent side 12. Find the hypotenuse to the nearest tenth.

16. A ladder leans against a wall. The ladder is 18 ft long and makes a 65-degree angle with the ground. How high up the wall does it reach?

Day 25 Problem Set

Continued | Mixed Synthesis

- | | |
|-----|---|
| 17. | A tree casts a 30 ft shadow when the angle of elevation to the sun is 42 degrees. Estimate the tree height. |
|-----|---|

- | | |
|-----|--|
| 18. | Two triangles are similar with scale factor 5. The smaller volume-like measurement is not applicable because triangles are 2D. What happens to area? |
|-----|--|

- | | |
|-----|--|
| 19. | A coordinate triangle has vertices (0,0), (6,0), and (6,8). Is it a right triangle? Explain. |
|-----|--|

- | | |
|-----|---|
| 20. | Spiral review: A circle has center (2,-3) and radius 5. Write its equation. |
|-----|---|

- | | |
|-----|---|
| 21. | Spiral review: A cylinder has radius 4 and height 10. Find its volume in terms of pi. |
|-----|---|

- | | |
|-----|--|
| 22. | Challenge: A ramp rises 3 ft over a horizontal distance of 12 ft. Find the angle the ramp makes with the ground to the nearest degree. |
|-----|--|